Input system:
Dependent variables:$r_1, z_2, z_1, z_3, x_1, y_3, x_3, p_1, q_2, q_3, p_3, x_2, q_1, y_1, y_2, r_3, p_2, r_2$
Independent variables:t
Equations:

$$\frac{dx_1}{dt} = p_1$$

$$\frac{dy_1}{dt} = r_1$$

$$\frac{dz_1}{dt} = q_1$$

$$\frac{dx_2}{dt} = p_2$$

$$\frac{dy_2}{dt} = r_2$$

$$\frac{dz_2}{dt} = q_2$$

$$\frac{dx_3}{dt} = p_3$$

$$\frac{dy_3}{dt} = r_3$$

$$\frac{dz_3}{dt} = q_3$$

$$\begin{aligned}\frac{dp_1}{dt} = {} & Gm_2 \left(x_2 - x_1\right) \operatorname{isqrt}\left(\left(x_1 - x_2\right)^2 + \left(y_1 - y_2\right)^2 + \left(z_1 - z_2\right)^2\right)^3 \\ & + Gm_3 \left(x_3 - x_1\right) \operatorname{isqrt}\left(\left(x_1 - x_3\right)^2 + \left(y_1 - y_3\right)^2 + \left(z_1 - z_3\right)^2\right)^3\end{aligned}$$

$$\begin{aligned}\frac{dr_1}{dt} = {} & Gm_2 \left(y_2 - y_1\right) \operatorname{isqrt}\left(\left(x_1 - x_2\right)^2 + \left(y_1 - y_2\right)^2 + \left(z_1 - z_2\right)^2\right)^3 \\ & + Gm_3 \left(y_3 - y_1\right) \operatorname{isqrt}\left(\left(x_1 - x_3\right)^2 + \left(y_1 - y_3\right)^2 + \left(z_1 - z_3\right)^2\right)^3\end{aligned}$$

$$\begin{aligned}\frac{dq_1}{dt} = {} & Gm_2 \left(z_2 - z_1\right) \operatorname{isqrt}\left(\left(x_1 - x_2\right)^2 + \left(y_1 - y_2\right)^2 + \left(z_1 - z_2\right)^2\right)^3 \\ & + Gm_3 \left(z_3 - z_1\right) \operatorname{isqrt}\left(\left(x_1 - x_3\right)^2 + \left(y_1 - y_3\right)^2 + \left(z_1 - z_3\right)^2\right)^3\end{aligned}$$

$$\frac{dp_2}{dt} = Gm_1 (x_1 - x_2) \operatorname{isqrt}\left((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2\right)^3 + Gm_3 (x_3 - x_2) \operatorname{isqrt}\left((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2\right)^3$$

$$\frac{dr_2}{dt} = Gm_1 (y_1 - y_2) \operatorname{isqrt}\left((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2\right)^3 + Gm_3 (y_3 - y_2) \operatorname{isqrt}\left((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2\right)^3$$

$$\frac{dq_2}{dt} = Gm_1 (z_1 - z_2) \operatorname{isqrt}\left((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2\right)^3 + Gm_3 (z_3 - z_2) \operatorname{isqrt}\left((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2\right)^3$$

$$\frac{dp_3}{dt} = Gm_1 (x_1 - x_3) \operatorname{isqrt}\left((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2\right)^3 + Gm_2 (x_2 - x_3) \operatorname{isqrt}\left((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2\right)^3$$

$$\frac{dr_3}{dt} = Gm_1 (y_1 - y_3) \operatorname{isqrt}\left((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2\right)^3 + Gm_2 (y_2 - y_3) \operatorname{isqrt}\left((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2\right)^3$$

$$\frac{dq_3}{dt} = Gm_1 (z_1 - z_3) \operatorname{isqrt}\left((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2\right)^3 + Gm_2 (z_2 - z_3) \operatorname{isqrt}\left((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2\right)^3$$

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Result:

Dependent variables:$r_1, z_3, p_1, x_3, q_2, q_3, s_2, p_3, q_1, r_3, p_2, s_1, z_2, z_1, x_1, y_3, x_2, y_1, y_2, s_3, r_2$

Independent variables:t

Additional variables:

$$s_1 = \operatorname{isqrt}\left((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2\right)$$

$$s_2 = \operatorname{isqrt}\left((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2\right)$$

$$s_3 = \operatorname{isqrt}\left((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2\right)$$

Equations:

$$\frac{dx_1}{dt} = p_1$$

$$\frac{dy_1}{dt} = r_1$$

$$\frac{dz_1}{dt} = q_1$$

$$\frac{dx_2}{dt} = p_2$$

$$\frac{dy_2}{dt} = r_2$$

$$\frac{dz_2}{dt} = q_2$$

$$\frac{dx_3}{dt} = p_3$$

$$\frac{dy_3}{dt} = r_3$$

$$\frac{dz_3}{dt} = q_3$$

$$\frac{dp_1}{dt} = s_1^3 G m_2 x_2 + (-1)\, s_1^3 G m_2 x_1 + s_2^3 G m_3 x_3 + (-1)\, s_2^3 G m_3 x_1$$

$$\frac{dr_1}{dt} = s_1^3 G m_2 y_2 + (-1)\, s_1^3 G m_2 y_1 + s_2^3 G m_3 y_3 + (-1)\, s_2^3 G m_3 y_1$$

$$\frac{dq_1}{dt} = s_1^3 G m_2 z_2 + (-1)\, s_1^3 G m_2 z_1 + s_2^3 G m_3 z_3 + (-1)\, s_2^3 G m_3 z_1$$

$$\frac{dp_2}{dt} = s_1^3 G m_1 x_1 + (-1)\, s_1^3 G m_1 x_2 + s_3^3 G m_3 x_3 + (-1)\, s_3^3 G m_3 x_2$$

$$\frac{dr_2}{dt} = s_1^3 G m_1 y_1 + (-1)\, s_1^3 G m_1 y_2 + s_3^3 G m_3 y_3 + (-1)\, s_3^3 G m_3 y_2$$

$$\frac{dq_2}{dt} = s_1^3 G m_1 z_1 + (-1)\, s_1^3 G m_1 z_2 + s_3^3 G m_3 z_3 + (-1)\, s_3^3 G m_3 z_2$$

$$\frac{dp_3}{dt} = s_2^3 G m_1 x_1 + (-1)\, s_2^3 G m_1 x_3 + s_3^3 G m_2 x_2 + (-1)\, s_3^3 G m_2 x_3$$

$$\frac{dr_3}{dt} = s_2^3 G m_1 y_1 + (-1)\, s_2^3 G m_1 y_3 + s_3^3 G m_2 y_2 + (-1)\, s_3^3 G m_2 y_3$$

$$\frac{dq_3}{dt} = s_2^3 G m_1 z_1 + (-1)\, s_2^3 G m_1 z_3 + s_3^3 G m_2 z_2 + (-1)\, s_3^3 G m_2 z_3$$

$$\frac{ds_1}{dt} = (-1)\, s_1^3 p_1 x_1 + s_1^3 p_1 x_2 + s_1^3 p_2 x_1 + (-1)\, s_1^3 p_2 x_2 + (-1)\, s_1^3 r_1 y_1 + s_1^3 r_1 y_2 + s_1^3 r_2 y_1 + (-1)\, s_1^3 r_2 y_2 + (-1)\, s_1^3 q_1 z_1 + s_1^3 q_1 z_2 + s_1^3 q_2 z_1 + (-1)\, s_1^3 q_2 z_2$$

$$\frac{ds_2}{dt} = (-1)\, s_2^3 p_1 x_1 + s_2^3 p_1 x_3 + s_2^3 p_3 x_1 + (-1)\, s_2^3 p_3 x_3 + (-1)\, s_2^3 r_1 y_1 + s_2^3 r_1 y_3 + s_2^3 r_3 y_1 + (-1)\, s_2^3 r_3 y_3 + (-1)\, s_2^3 q_1 z_1 + s_2^3 q_1 z_3 + s_2^3 q_3 z_1 + (-1)\, s_2^3 q_3 z_3$$

$$\frac{ds_3}{dt} = (-1)\, s_3^3 p_2 x_2 + s_3^3 p_2 x_3 + s_3^3 p_3 x_2 + (-1)\, s_3^3 p_3 x_3 + (-1)\, s_3^3 r_2 y_2 + s_3^3 r_2 y_3 + s_3^3 r_3 y_2 + (-1)\, s_3^3 r_3 y_3 + (-1)\, s_3^3 q_2 z_2 + s_3^3 q_2 z_3 + s_3^3 q_3 z_2 + (-1)\, s_3^3 q_3 z_3$$